

Homework 4

JOLDESCU CRINA-MIHAIELA
Group 914

① Find the radius of convergence and the convergence set for each of the following series.

* Power Series: $\sum_{n \geq 0} a_n (x-c)^n$

$$a) \sum_{n \geq 0} \frac{n x^n}{2^n} = \begin{cases} a_n = \frac{n}{2^n} \\ c = 0 \end{cases}$$

$$= \sum_{n \geq 0} \frac{n}{2^n} \cdot x^n$$

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n}{2^n}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n}}{\underbrace{\sqrt[n]{2^n}}_{=2}} = \frac{1}{2} \in (0, +\infty) \Rightarrow$$

$$\Rightarrow R = \frac{1}{L} = \frac{1}{\frac{1}{2}} = 2 \Rightarrow \text{the series is absolutely convergent for } x \in (-2, 2)$$

R=2

Check the endpoints:

$$\text{I } x = -2 \Rightarrow \sum_{n \geq 0} \frac{n \cdot (-2)^n}{2^n} = \sum_{n \geq 0} n \cdot (-1)^n \rightarrow \text{diverges}$$

$$\text{II } x = 2 \Rightarrow \sum_{n \geq 0} \frac{n \cdot 2^n}{2^n} = \sum_{n \geq 0} n \rightarrow \text{diverges}$$

$$\Rightarrow \boxed{C = (-2, 2)}$$

$$b) \sum_{n \geq 1} \frac{x^{2n}}{\sqrt{n}} = \sum_{n \geq 1} \frac{(x^2)^n}{\sqrt{n}}$$

$$\text{Let } y = x^2$$

$$\Rightarrow \sum_{n \geq 1} \frac{x^{2n}}{\sqrt{n}} = \sum_{n \geq 1} \frac{y^n}{\sqrt{n}} \quad \begin{cases} a_n = \frac{1}{\sqrt{n}} \\ c = 0 \end{cases}$$

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{\sqrt{n}}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n^{\frac{1}{2}}}} =$$

$$= \lim_{n \rightarrow \infty} n^{-\frac{1}{2n}} = \lim_{n \rightarrow \infty} e^{\ln(n^{-\frac{1}{2n}})} = \lim_{n \rightarrow \infty} e^{\frac{-1}{2n} \cdot \ln n} \quad (*)$$

$$\textcircled{*} \lim_{n \rightarrow \infty} \frac{-1}{2n} \cdot \ln n = - \lim_{n \rightarrow \infty} \frac{\ln n}{2n} \stackrel{\frac{\infty}{\infty}}{=} \frac{0}{1} = 0$$

$$= - \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{2} = - \lim_{n \rightarrow \infty} \frac{1}{2n} = - \frac{1}{\infty} = 0$$

$$\Rightarrow L = e^0 = 1 \in (0, +\infty)$$

$$\Rightarrow R = \frac{1}{L} = 1 \Rightarrow \text{the series is absolutely convergent for } x \in (-1, 1)$$

$$\boxed{R=1}$$

Check the endpoints:

$$|x^2| < 1 \Rightarrow |x| < 1$$

$$\text{I } x = -1 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} < 1 \quad x \in (-1, 1)$$

$\Downarrow$ diverges

$$\text{II } x = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} < 1 \Rightarrow \text{diverges}$$

$$\Rightarrow \boxed{C = (-1, 1)}$$

$$\textcircled{c) } \sum_{n=1}^{\infty} (-1)^n \frac{(x-1)^n}{n} \quad \left\{ \begin{array}{l} a_n = \frac{(-1)^n}{n} \\ c = 1 \end{array} \right.$$

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{(-1)^n}{n}} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n}} = 1 \in (0, +\infty)$$

$$\Rightarrow R = \frac{1}{L} = 1 \quad \boxed{R=1} \Rightarrow \text{the series is absolutely convergent for } x \in (0, 2)$$

Check the endpoints:

$$\text{I } x = 0 \Rightarrow \sum_{n=1}^{\infty} (-1)^n \cdot \frac{(-1)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow \text{diverges}$$

$$\text{II } x = 2 \Rightarrow \sum_{n=1}^{\infty} (-1)^n \cdot \frac{1^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \xrightarrow[\text{Text}]{\text{Leibniz}} \text{converges}$$

$$\Rightarrow \boxed{C = (0, 2]}$$

② Study the convergence and compute the sum for the series

$$\sum_{n \geq 2} \frac{x^n}{n(n-1)}.$$

$$\begin{cases} a_n = \frac{1}{n(n-1)} \\ c = 0 \end{cases}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{1}{(n+1)n} \cdot \frac{n(n-1)}{1} = \lim_{n \rightarrow \infty} \frac{n-1}{n+1} = 1 \in (0, +\infty) \Rightarrow$$

$\Rightarrow R = \frac{1}{L} = \frac{1}{1} = 1 \Rightarrow$ the series is absolutely convergent for $x \in (-1, 1)$, that is for $|x| < 1$

$$\begin{aligned} \sum_{n \geq 2} \frac{x^n}{n(n-1)} &= \sum_{n \geq 2} \frac{1}{n(n-1)} \cdot x^n = \sum_{n \geq 2} \left(\frac{1}{n-1} - \frac{1}{n} \right) x^n = \\ &= \sum_{n \geq 2} \frac{x^n}{n-1} - \sum_{n \geq 2} \frac{x^n}{n} = \sum_{k \geq 1} \frac{x^{k+1}}{k} - \sum_{n \geq 2} \frac{x^n}{n} = \\ &= x \cdot \sum_{k \geq 1} \frac{x^k}{k} - \left(\sum_{n \geq 1} \frac{x^n}{n} - x \right) = \end{aligned}$$

• we know that $\sum_{n \geq 1} \frac{x^n}{n} = -\ln(1-x)$, for $|x| < 1$
(obtained by integrating the geometric series) •

$$\begin{aligned} &= x(-\ln(1-x)) - (-\ln(1-x) - x) = \\ &= -x \cdot \ln(1-x) + \ln(1-x) + x = \\ &= (1-x) \ln(1-x) + x \end{aligned}$$

Conclusion: $\rightarrow \sum_{n \geq 2} \frac{x^n}{n(n-1)}$ is convergent for $|x| < 1$ and

$$\sum_{n \geq 2} \frac{x^n}{n(n-1)} = (1-x) \ln(1-x) + x$$

$\rightarrow \sum_{n \geq 2} \frac{x^n}{n(n-1)}$ is divergent for $|x| > 1$

Check the endpoints

$$I \quad x = -1 \Rightarrow \sum_{n \geq 2} \frac{(-1)^n}{n(n-1)} = \sum_{n \geq 2} \left(\frac{(-1)^n}{n-1} - \frac{(-1)^n}{n} \right) =$$

$$= \sum_{n \geq 2} \frac{(-1)^n}{n-1} - \sum_{n \geq 2} \frac{(-1)^n}{n} =$$

$$(m = k+1)$$

$$= \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} - \left(\sum_{n \geq 1} \frac{(-1)^n}{n} - \frac{(-1)^1}{1} \right) =$$

$$= \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} - \sum_{n \geq 1} \frac{(-1)^n}{n} + 1 =$$

$$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \ln 2 \Rightarrow \sum_{n \geq 1} \frac{(-1)^n}{n} = -\ln 2$$

(alternating harmonic series)

$$= \ln 2 - (-\ln 2) - 1 = 2 \ln 2 - 1$$

$\Rightarrow$ the series converges and its sum is $2 \ln 2 - 1$

$$II \quad x = 1 \Rightarrow \sum_{n \geq 2} \frac{1}{n(n-1)} = \sum_{n \geq 2} \left(\frac{1}{n-1} - \frac{1}{n} \right) =$$

$$= \lim_{n \rightarrow \infty} \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n-1} - \frac{1}{n} \stackrel{\lim}{\rightarrow} 1 - \frac{1}{n} = 1$$

$\Rightarrow$ the series converges and its sum is 1

hw4.py ×

```
1 import math
2
3 def alt_harmonic_partial(n): 1 usage
4     # partial sum of the alternating harmonic series up to n terms
5     return sum((-1)**(k+1) / k for k in range(1, n+1))
6
7 def rearranged_sum(p, q, total_terms=1000000): 1 usage
8     # rearranged version: add p positive terms, then q negative terms, and so on...
9     positives = [1 / n for n in range(1, total_terms*2, 2)] #1, 1/3, 1/5....
10    negatives = [-1 / n for n in range(2, total_terms*2+1, 2)] #-1/2, -1/4, -1/6....
11
12    s = 0 # start with sum 0
13    pos_index = 0
14    neg_index = 0
15    results = []
```

```
16
17    while pos_index < len(positives) and neg_index < len(negatives):
18        # add p positive terms
19        for _ in range(p):
20            if pos_index >= len(positives):
21                break
22            s += positives[pos_index]
23            pos_index += 1
24            results.append(s)
25        # add q negative terms
26        for _ in range(q):
27            if neg_index >= len(negatives):
28                break
29            s += negatives[neg_index]
30            neg_index += 1
31            results.append(s)
32
33    return results[-1] # final sum after rearrangement
34
```

```
35
36 def main(): 1 usage
37     true_ln2 = math.log(2)
38     print("ln(2) =", true_ln2)
39     print()
40
41     print("Partial sums:")
42     print("-" * 50)
43
44     for n in [1, 2, 5, 10, 50, 200, 1000, 5000]:
45         s = alt_harmonic_partial(n)
46         print("n =", n, "; partial sum =", s, "; error =", s - true_ln2)
47         # as n grows, the error should become smaller and smaller
48
49     print()
50     print("Rearranged series results:")
51     print("-" * 50)
52
```

```
53
54     # (p,q) is the "pattern" for the series: we add p positive numbers, we add q negative numbers, we then
55     # add p positive numbers again and so on...
56     for (p, q) in [(1, 1), (2, 1), (3, 2), (5, 2), (10, 3)]:
57         final_sum = rearranged_sum(p, q)
58         print("(p =", p, ", q =", q, ") → final sum ≈", final_sum)
59
60
61 if __name__ == "__main__":
62     main()
```

```
Run hw4 x
C:\Users\Crina\PycharmProjects\PythonProjectHW_MA\.venv\Scripts\python.exe C:\Users\Crina\PycharmProjects\PythonProjectHW_MA\hw4.py
ln(2) = 0.6931471805599453

Partial sums:
-----
n = 1 ; partial sum = 1.0 ; error = 0.3068528194400547
n = 2 ; partial sum = 0.5 ; error = -0.1931471805599453
n = 5 ; partial sum = 0.7833333333333333 ; error = 0.09018615277338804
n = 10 ; partial sum = 0.6456349206349206 ; error = -0.047512259925024725
n = 50 ; partial sum = 0.6832471605759182 ; error = -0.00990001998402712
n = 200 ; partial sum = 0.6906534304818243 ; error = -0.0024937500781210264
n = 1000 ; partial sum = 0.6926474305598203 ; error = -0.0004997500001250321
n = 5000 ; partial sum = 0.6930471905599451 ; error = -9.99900000016078e-05

Rearranged series results:
-----
(p = 1 , q = 1 ) → final sum ≈ 0.6931469305601015
(p = 2 , q = 1 ) → final sum ≈ 1.0397202708399997
(p = 3 , q = 2 ) → final sum ≈ 0.8958783596159483
(p = 5 , q = 2 ) → final sum ≈ 1.1512919214971817
(p = 10 , q = 3 ) → final sum ≈ 1.2951327493899294

Process finished with exit code 0
```